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SCHOOL FUTURE TECH PROGRAMM

Faculty Name :

Course Duration	: 5 Months (150 Days)
Classes	: Offline & Online
Daily Teaching	: 3 Hours
Total Hours	: 300 Hours

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LEVEL 1 – JUNIOR INNOVATORS (5th–6th Standard)

MODULE 1 – COMPUTER BASICS (Classes 1–20)

Class 8 — Operating Systems – Windows, Android & Linux (Complete Notes)

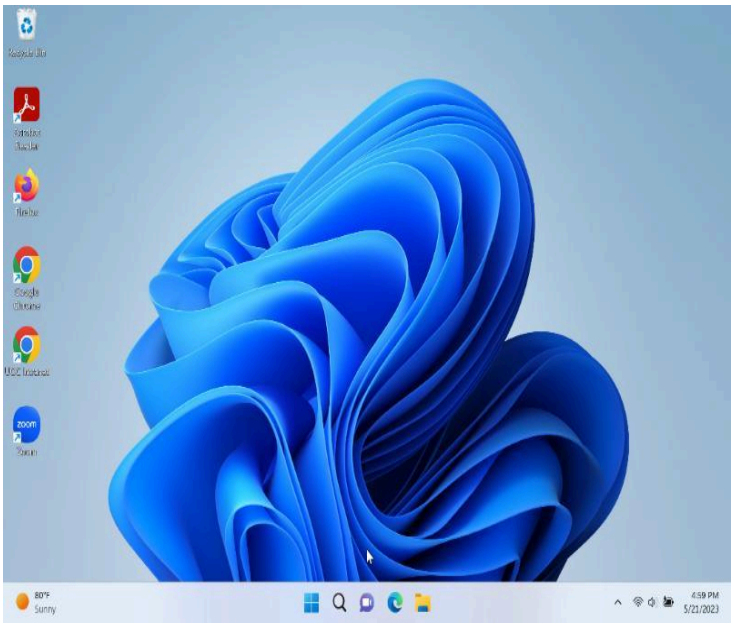
Complete Theory & Practical Notes

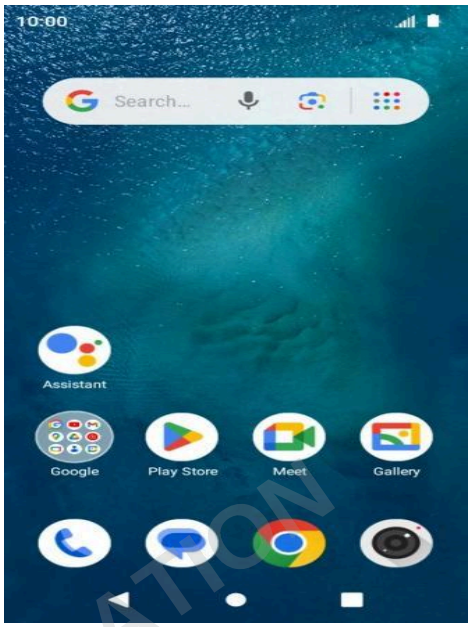
Learning Objectives

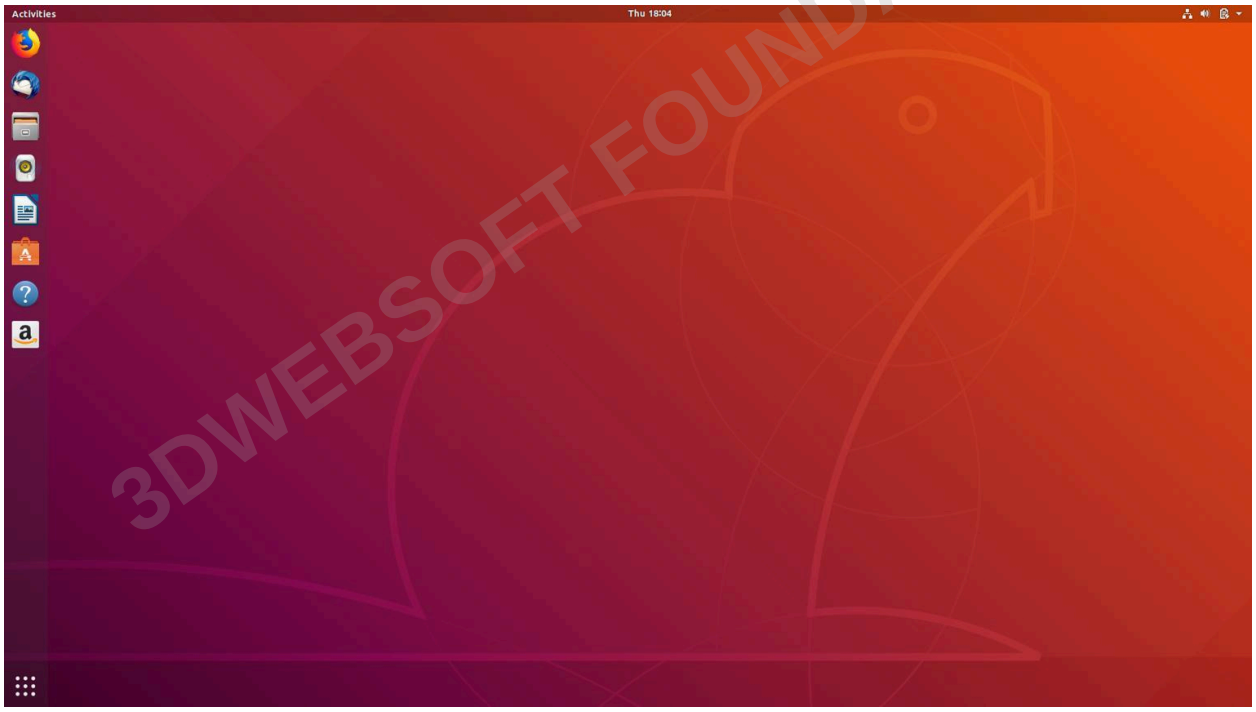
By the end of this class, students will be able to:

- Understand what an Operating System (OS) is.
- Explain why an Operating System is important.
- Identify the features of Windows, Android, and Linux.
- Compare different operating systems.
- Perform basic operating system activities.
- Create an interactive Scratch project about Operating Systems.

What is an Operating System (OS)?







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Definition:-

An **Operating System (OS)** is **System Software** that manages the computer and allows users to interact with it.

The Operating System controls the computer's hardware, runs applications, manages files, and helps users perform tasks.

Simple Example

Think of the Operating System as the ***Principal of a School.***

- The Principal manages teachers and students.
- The Operating System manages the computer's hardware and software.

Without an Operating System, a computer or mobile phone cannot work properly.

Functions of an Operating System

The Operating System performs many important tasks.

Main Functions:-

Starts the computer (Booting)

Runs applications

Manages files and folders

Controls keyboard, mouse, printer, and other devices

Manages memory

Provides security

Helps users interact with the computer

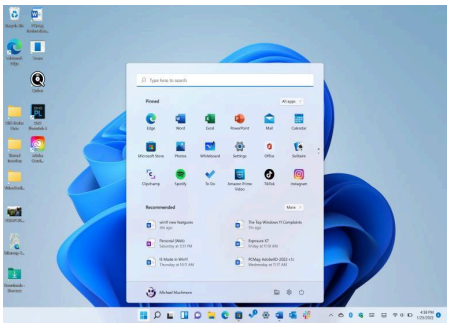
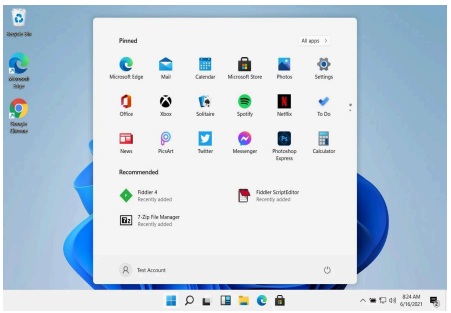
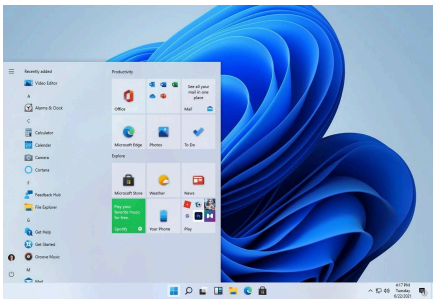
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Types of Operating Systems

Today we will learn about three popular Operating Systems:-

1. Windows
2. Android
3. Linux

1. Windows Operating System

What is Windows?

Windows is one of the most popular Operating Systems developed by Microsoft.

It is mainly used on desktop computers and laptops.

Features

- Easy to use
- Colorful graphical interface
- Supports many software applications
- Good for schools, homes, and offices

Uses

- Typing documents
- Internet browsing
- Programming
- Online learning
- Playing games

Examples

- Windows 10
- Windows 11

2. Android Operating System

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What is Android?

Android is an Operating System mainly used in smartphones and tablets.

It allows users to install apps, browse the internet, take photos, and play games.

Features

- Touchscreen interface
- App Store support
- Voice Assistant

- Wi-Fi and Bluetooth support

Uses

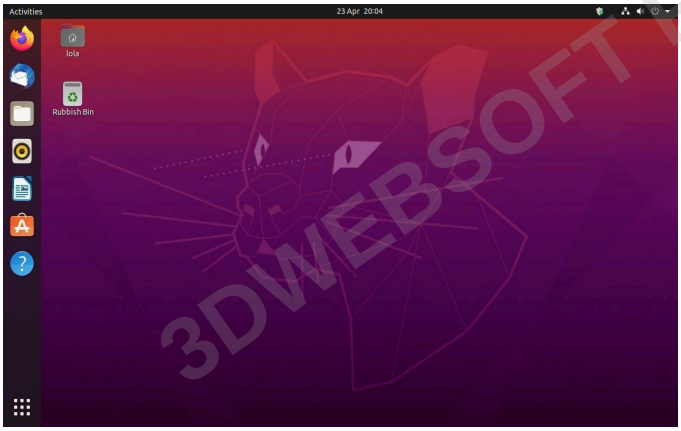
- Calling
- Messaging
- Online classes
- Photography
- Mobile gaming

Examples

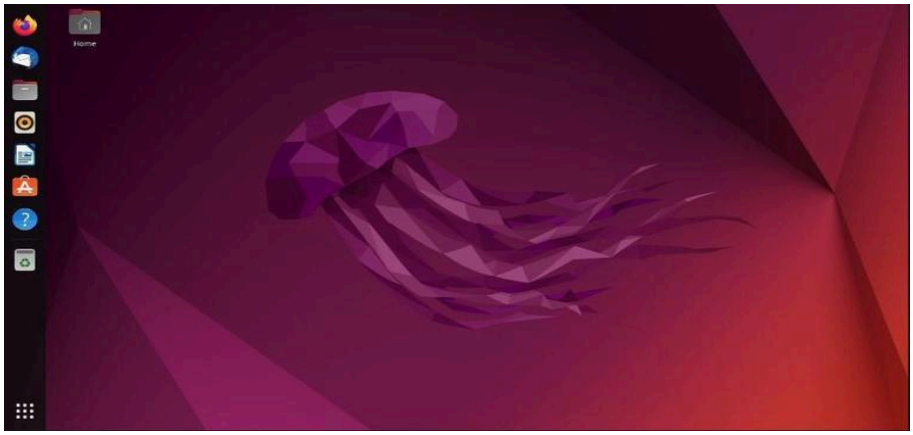
- Samsung Galaxy
- Google Pixel
- OnePlus
- Xiaomi

3. Linux Operating System

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What is Linux?

Linux is a free and open-source Operating System.

It is widely used in servers, schools, software development, and supercomputers.

Features

- Free to use
- Secure
- Fast
- Reliable
- Customizable

Uses

- Programming
- Web servers
- Artificial Intelligence
- Cloud Computing
- Scientific Research

Popular Linux Versions

- Ubuntu
- Fedora
- Linux Mint

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Comparison of Operating Systems

Feature	Windows	Android	Linux
Device	Computer	Mobile & Tablet	Computer & Server
Touch Support	Limited	Excellent	Depends on Version
Free	No	Mostly Yes	Yes
Easy for Beginners	Yes	Yes	Moderate
Used For	Home, School, Office	Mobile Apps	Programming & Servers

Operating Systems by Common Device

Common Device for Each Operating System

Typical devices where each operating system is commonly used.

036912Windows (PC)Android (Mobile)Linux (Server/PC)

Easy Memory Trick

Operating System	Remember Like
Windows	Computer Expert
Android	Mobile Friend
Linux	Programmer's Choice

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Classroom Practical Activity

Activity 1 – Identify the Operating System

The teacher shows pictures of different devices.

Students identify:

- Which Operating System is used?
- Is it a Computer or Mobile?
- Where is it commonly used?

Example:-

Device	Operating System
Laptop	Windows
Smartphone	Android
School Computer Lab	Windows
Programming Computer	Linux

Activity 2 – Explore the Desktop

Students perform simple tasks:

Windows

- Open the Start Menu.
- Open Paint or Calculator.
- Close the application.

Android

- Open the Camera app.
- Open Settings.
- Adjust screen brightness.

Linux (Teacher Demonstration)

- Show the desktop.
- Open the File Manager.
- Open a web browser.

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Activity 3 – Match the Operating Syste

Operating System	Used In
Windows	Desktop & Laptop
Android	Mobile Phone & Tablet
Linux	Server & Programming Computer

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Practical Coding Activity (Scratch)

Project Title

"Operating System Explorer"

Objective

Students create an interactive Scratch project where clicking an operating system displays its name and purpose.

Step 1

Open **Scratch.**

Create a new project.

Delete the default Cat sprite.

Step 2

Add Sprites

- Windows Logo
- Android Robot

- Linux Penguin

Choose a computer classroom background.

Step 3 – Programming

Green Flag

when green flag clicked
say "Welcome! Click an Operating System to learn about it." for 3 seconds

Windows Sprite

when this sprite clicked
say "I am Windows. I help computers run applications." for 4 seconds

Android Sprite

when this sprite clicked
say "I am Android. I power smartphones and tablets." for 4 seconds

Linux Sprite

when this sprite clicked
say "I am Linux. I am used to programming, servers, and AI." for 4 seconds

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Advanced Scratch Challenge

Create a quiz with a ***Score*** variable.

Question 1

ask "Which Operating System is mainly used on mobile phones?" and wait

If answer = Android

Then

change Score by 1
say "Correct!" for 2 seconds

Else

say "Try Again!" for 2 seconds

Add More Questions

- Which Operating System is popular on desktop computers?
- Which Operating System is free and open-source?
- Which Operating System is widely used for servers?
- Which Operating System is developed by Microsoft?

Display the total score at the end.

AI Connection

Discuss with students:-

- Which Operating System does your phone use?
- Which Operating System is commonly used for AI and cloud servers? (Linux)
- Can AI applications run on Windows and Android? (Yes)
- Why do software developers often use Linux?

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Fun Quiz:-

1. What is an Operating System?

- A) A keyboard
- **B) Software that manages a computer**
- C) A printer

2. Which Operating System is commonly used on laptops?

- **A) Windows**
- B) Android
- C) Memory Card

3. Which Operating System is mainly used on smartphones?

- A) Linux
- **B) Android**
- C) HDD

4. Which Operating System is free and open-source?

- A) Windows

- B) Android
- **C) Linux**

5. Which company develops Windows?

- A) Google
- **B) Microsoft**
- C) Apple

Real-Life Activity:-

Ask students to identify the operating systems used at home:-

Device	***Operating System***
Family Laptop	*Windows (or another OS)*
Smartphone	*Android (or another OS)*
School Computer	*Windows*
Computer Lab	*Windows or Linux*

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Homework:-

1. Draw the logos of ***Windows, Android, and Linux.***
2. Write ***three features*** of each Operating System.
3. Ask family members which Operating System they use on their devices.
4. Create a comparison table of ***Windows, Android, and Linux.***
5. ***Improve your Scratch project by adding:-***
    - Sound effects
    - Animations
    - A score counter
    - A "Congratulations!" message when the quiz is completed.

Learning Outcomes:-

After completing this lesson, students will be able to:

- Define an ***Operating System*** and explain its importance.

- Identify the features and uses of **Windows, Android, and Linux.**
- Compare different operating systems and the devices they run on.
- Perform simple tasks using common operating systems.
- Build an interactive Scratch project to teach others about operating systems.
- Understand how operating systems help computers and mobile devices run software and interact with users.

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